

Transshipment problem

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Practice quiz

Consider three factories which produce honey in Boston, New York and Los Angeles, whose supplies are 200 units, 250 units and 300 units per day, respectively. Customers have a demand of 250 units per day in each city. The costs for transporting one unit are the following:

- from Los Angeles to Boston is \$1,
- from Los Angeles to New York is \$0.5,
- from New York to Boston is \$0.8.

A maximum of 40 units per day can be transported from one city to another. Moreover, due to a specific contract with the shipping company, we have to transport at least 20 units from New-York to Boston. We want to find the cheapest way to transport the honey from the factories to the customers.

1. Represent the problem as a network. Notify the supply/demand data s_i next to each node i . Next to each arc (i, j) , notify the cost and the capacities using the notation $c_{ij}(\ell_{ij}, u_{ij})$.
2. Write the transshipment problem as a linear optimization problem.
3. Represent the network that corresponds to the standard form formulation.
4. Identify the data $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ of the transshipment problem in standard form:

$$\min_{x \in \mathbb{R}^n} c^T x$$

subject to

$$\begin{aligned}Ax &= b, \\ x &\geq 0.\end{aligned}$$

5. What is the rank of the matrix A ?